

MODULE 4C — Email CRM & Communication Engine

Parent Module: Module 4 — AI Calling Engine

SRS v1.0 | Status: Draft → Pending Confirmation

Category: Omni-Channel Engagement (Conversational CRM)

1. Purpose

This sub-module enables **Email as a first-class CRM engagement channel**, tightly integrated with:

- Lead Management (Module 3)
- AI Calling Engine (Module 4)
- WhatsApp CRM (Module 4A)
- SMS CRM (Module 4B)
- User & Role Management (Module 2)
- Tenant & License Management (Module 1)

Email is positioned as:

- A high-context, long-form communication channel
 - A campaign and nurture backbone
 - A compliance-friendly audit medium
 - A bridge between automation and human interaction
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2. Scope

2.1 In Scope

- Inbound & outbound email handling
- Lead creation & enrichment from email
- AI-assisted email drafting & replies
- Human-driven email conversations
- Campaign-based email sending
- Template & signature management
- Threading & reply tracking
- CRM timeline integration
- Email server / provider configuration UI

- Email deliverability analytics (SPF/DKIM dashboards)
 - External email CRM synchronization
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3. Email Channel Model

3.1 Sender Identity

- Platform-managed sending domains
- Tenant-scoped sender addresses (e.g., support@, sales@)
- Optional user-based sender aliases (license-controlled)
- No BYO SMTP credentials in v1

All sender identities are:

- Verified
 - Audited
 - License-governed
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3.2 Conversation & Thread Model

Each email thread must be associated with:

- Tenant ID
- Lead ID (existing or auto-created)
- Assigned owner (user or AI)
- Campaign ID (if applicable)
- Thread ID (immutable)

Email threads are treated as **stateful conversations**, not isolated messages.

4. Email Lead Capture & CRM Binding

4.1 Inbound Email Handling

On inbound email:

- Match lead via sender email
- If not found → auto-create lead

- Source = Email
- First-touch timestamp recorded

Captured metadata:

- Sender email
 - Subject
 - Body (HTML + text)
 - Attachments
 - Timestamp
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4.2 Email-Based Lead Actions

Emails may trigger:

- Lead stage updates
- Task creation
- Callback scheduling
- Campaign enrollment
- AI follow-up calls

Triggers must be rule-based and auditable.

5. Email Messaging Capabilities

5.1 Supported Content

- Plain text
 - HTML emails
 - Inline images
 - File attachments
 - Tracking links
 - Email signatures
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5.2 Personalization Tokens

Templates may include:

- Lead name

- Company name
- Assigned agent
- Appointment details
- Dynamic CTA links

Token validation required before send.

6. AI-Powered Email Assistance

6.1 AI Capabilities

AI is allowed to:

- Draft outbound emails
- Suggest replies
- Summarize inbound threads
- Tag sentiment & intent
- Recommend next actions
- Trigger cross-channel follow-ups

AI is **not allowed** to:

- Send bulk emails without approval
 - Bypass opt-out rules
 - Modify compliance-sensitive fields
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6.2 AI Context Sources

AI consumes:

- Lead profile & history
- Past emails & threads
- Call transcripts
- WhatsApp/SMS interactions
- Campaign objectives
- Tenant compliance rules

Context injection must be logged.

7. Human + AI Collaboration Model

7.1 Ownership Modes

- AI-drafted, human-approved
- Human-authored
- Fully AI-managed (transactional use cases)

Ownership must be explicit per message.

7.2 Supervisor Controls

Managers may:

- Review drafts
 - Enforce approvals
 - Reassign threads
 - Freeze AI sending
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8. Email Campaigns

8.1 Campaign Types

- One-time broadcasts
 - Drip/nurture campaigns
 - Event-triggered emails
 - Transactional notifications
 - Call-to-email campaigns
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8.2 Campaign Controls

- Template approval enforcement
- Opt-in & unsubscribe handling
- Quiet hours (configurable)
- Per-campaign send caps
- Cost-aware throttling

Campaign interactions must update:

- Lead timeline
 - Engagement score
 - Lead stage (where applicable)
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9. Template & Signature Management

9.1 Template Lifecycle

- Draft
- Approved
- Active
- Archived

Templates include:

- Subject
 - Body (HTML/text)
 - Variables
 - Language
 - Category (marketing, transactional, support)
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9.2 Signatures

- Tenant-level signatures
 - Role-based signatures
 - User-specific signatures (optional)
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10. Automation & Workflow Triggers

Email events may trigger:

- Lead stage changes
- Task creation
- SLA escalation
- AI call initiation
- WhatsApp/SMS fallback
- Campaign branching

Automations must be deterministic and logged.

11. Activity Timeline Integration

Every email event must appear in the **Lead Timeline (Module 3)**:

- Sent emails
- Received emails
- Attachments
- AI actions
- Human actions
- Campaign interactions

Timeline is immutable and chronological.

12. Search, Filters & Views

Users can search/filter by:

- Subject
- Sender / recipient
- Lead stage
- Campaign
- Unread status
- Sentiment (AI-tagged)
- Attachment presence

Saved views supported.

13. Permissions & Security

13.1 Role-Based Permissions

Permissions include:

- View email threads
- Draft emails
- Send emails

- Approve templates
- Export emails (if allowed)

Controlled via **Module 2 RBAC**.

13.2 Compliance & Privacy

- Unsubscribe handling mandatory
 - Email suppression lists enforced
 - Data retention policies applied
 - Attachment scanning supported (optional)
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14. Cost Control & Safeguards

- Daily send limits
- Per-campaign caps
- AI draft throttling
- Spam-risk prevention rules
- Forced suspension on abuse detection

No uncontrolled email volume allowed.

15. Reporting Hooks (Data Emission Only)

Email module emits:

- Delivery & open events
- Click-through actions
- Reply rates
- Conversion triggers
- AI vs human performance

Dashboards handled in **Module 7**.
